

Visualisation Packages

package	Language	extra_description	Comment
matplotlib	python		The base in python. Works.
plotly express	python		Generate interactive html views. QoL features like continuous color scales managed automatically
scikit-optimize	python	Visualise ML optimisation results	I didn't test the build-in options.
seaborn	python	matplotlib-associated	Works well with Matplotlib. Better ergonomics. The reference for Python visualisation.
base	R		Excellent for simple graphs, but cumbersome for more complicated visualisations.
ggnewscale	R	ggplot-associated	Creates an infinite number of new scales, allowing to have several colors or fill in the same plot.
ggplot2	R		The reference for R plotting. The syntax needs to be studied but is remarkably understandable when we get it. Has MANY associated packages to expand its versatility almost indefinitely.
ggpubr	R	ggplot-associated	Great overall package to make publication-level plots
ggtree	R	ggplot-associated	Excelptional visualization tool for phylogenetic trees.
ggtreeExtra	R	ggplot-associated	Extra tools for ggtree
ggvenndiagram	R		Better Venn diagrams than the original ggplot
gridExtra	R	ggplot-associated	Gives more flexibility for composite plots
kable	R		Tool to format clean, publishable tables.

Language

Tested

Standalone Tools

Name	Description	Tags	Comment	Tested
Dbeaver	DBeaver is free and open source universal database tool for developers and database administrators.	Database Management, SQL	Supports several SQL versions. Good and clear UI and QoL.	y
EDrawMind	A collaborative mind mapping, brainstorming, and outlining tool	Mind Mapping	Clean and efficient mind mapping tool. Perpetual license available.	y
Geneious	Geneious has all the molecular biology and sequence analysis tools your organization needs. Subscription-based.	Biology, Genetics, Visualisation	Excellent software to handle phylogenetic sequences, create and annotate phylogenetic trees. Ergonomic is excellent and has a rich documentation. Read- and export to many trees extensions.	y
KNIME	KNIME is a single platform for end-to-end data science. Beyond building workflows, commercial teams leverage KNIME to ensure sensitive data is kept safe, analytics and AI models are validated and monitored, and results are verified and explainable.	Data Analysis, Visualisation, No programming	The basic functionalities are easy to understand, particularly interesting tool for people who do not know programming. The chart creation has limitations but is probably good enough for most use cases.	y
Looping	Looping est un logiciel de modélisation conceptuelle de données entièrement gratuit et libre d'utilisation.	Data modelling	Great data modelling tool. Has many options and a lot of flexibility, with the possibility to use several colors, add notes, edit links...	y

Machine Learning methods

Package

Type

box_type

Tested ML methods

Function	Type	box_type	Tested	test_dataset	Comments
BayesianSearch	optimisation		yes	forgery	2-step process with early stop give excellent results
GridSearch	optimisation		yes	forgery	Works but requires at least 2 rounds, long if there are many parameters.
Random Forest Classifier	supervised	blackbox	yes	forgery	Fast, accurate
xgb	supervised	blackbox	yes	forgery	Fast, accurate
K Nearest Neighbors	supervised	blackbox	yes	forgery	Instant training, accurate results
Random Forest Regressor	supervised	blackbox	yes	forgery	More accurate than Linear regression with forgery data.
Gaussian Mixture Model	unsupervised	blackbox	yes	forgery	Very fast training, the most accurate of all the algorithms tested
Linear Regression	supervised	glassbox	yes	forgery	Inaccurate due to unadapted data.
Logistic regression	supervised	glassbox	yes	forgery	Somewhat slower, accurate
Explainable Boosting Classifier	supervised	glassbox	yes	forgery	Very slow, less precise than the other ones, but great interpretability interface
K-Means	unsupervised	blackbox	yes	market_study	Fast and relevant results
Gaussian Mixture Model	unsupervised	blackbox	yes	market_study	Fast and relevant results, allows clusters with an oval shape while they stay "round" with K-means
Principal Component Analysis	unsupervised	blackbox	yes	market_study	Relevant results
Random Search	optimisation		no		Could be an interesting first step instead of Bayesian Search
HDBSCAN	unsupervised	blackbox	no		Clustering based on the distance: Allow very long or circular clusters.
LIME	explainer	blackbox	no		Explainer method for blackbox machine learning algorithms

6

Untested Methods

P13 Project planning

